

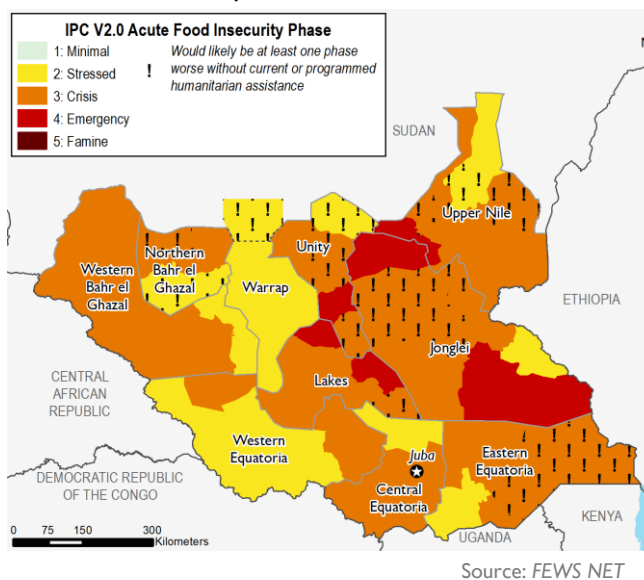
Famine (IPC Phase 5) risk persists and substantial scale-up of assistance needed

FEWS NET produces forward-looking food security analysis and IPC compatible mapping several times a year for 28 countries, including South Sudan. FEWS NET is a member of South Sudan's multi-stakeholder IPC Technical Working Group and an active participant in national IPC analysis workshops in South Sudan. The map and classifications in this report use IPC standards and methods, but do not necessarily reflect a consensus view of the national IPC Technical Working Group, IPC partners, or the Government of the Republic of South Sudan. However, this report was reviewed by the South Sudan IPC Technical Working Group prior to publication. The next IPC analysis will take place in January/February 2019.

KEY MESSAGES

- During the peak of the 2018 lean season, over 6 million people were estimated to have been in Crisis (IPC Phase 3) or worse, including over 1.7 million people in Emergency (IPC Phase 4) and 47,000 in Catastrophe (IPC Phase 5). This was the need in the presence of humanitarian assistance.
- Food security is expected to improve through the end of 2018, though some households did not harvest and will not share in improvements. Extreme levels of acute food insecurity will continue throughout the projection period, and some households will likely be in Catastrophe (IPC Phase 5)¹ in Leer, Mayendit, Greater Baggari, Yirol West, Yirol East, Pigi/Canal, and Panyikang.
- Humanitarian food assistance continues to play a key role in reducing the occurrence of extreme food insecurity. However, planned assistance is estimated to reach only 2-3 million people a month throughout the outlook period, far below the need. Emergency (IPC Phase 4), and associated excess mortality, is likely in several areas in the absence of higher levels of food assistance and unhindered access to deliver assistance.
- In Greater Baggari, a September MUAC screening indicated outcomes had deteriorated sharply since mid-2018, from which time humanitarians have faced significant access constraints and minimal assistance has been delivered. Should conflict continue to restrict household movement and humanitarian access, Famine (IPC Phase 5) would be possible. In central Unity, data indicated that many households were in Catastrophe (IPC Phase 5) during the lean season. With the recent return of humanitarians to the area, consumption has improved slightly, though given the expectation of little to no harvests among households of concern, Famine (IPC Phase 5) is likely should conflict restrict the delivery of assistance throughout the projection period.
- Although Greater Baggari and central Unity are of greatest concern currently, new areas of concern can arise quickly when an uptick in violence restricts household movement and humanitarian access, which can in turn result in significant food security deterioration. An end to the conflict, and full implementation of the peace deal is ultimately needed to end the persistent risk of Famine (IPC Phase 5) in South Sudan.

Current food security outcomes, October 2018



FEWS NET classification is IPC-compatible. IPC-compatible analysis follows key IPC protocols but does not necessarily reflect the consensus of national food security partners.

¹ According to the IPC, a Famine (IPC Phase 5) has occurred when at least 20 percent of households in a given area have an extreme lack of food, the Global Acute Malnutrition (GAM) prevalence, as measured by weight-for-height z-score (WHZ), exceeds 30 percent, and mortality, as measured by the Crude Death Rate (CDR), is greater than 2 per 10,000 per day. Catastrophe (IPC Phase 5) is when a household group has an extreme lack of food and/or other basic needs even with the full employment of coping strategies.

South Sudan to continue facing Famine (IPC Phase 5) risk despite peace deal

Data on food consumption, nutrition, and mortality collected across South Sudan between May and August 2018, the peak of the lean season, pointed to extreme outcomes, indicative of Emergency (IPC Phase 4) or worse, across the country (Figure 1). According to August 2018 Food Security and Nutrition Monitoring System (FSNMS) data, outcomes are broadly similar to the 2017 lean season, though notable improvements were observed in some areas, while deterioration was recorded in others.

In September, the IPC Technical Working Group estimated [6.1 million people](#)² (59% of the population) were in Crisis (IPC Phase 3) or worse, of whom 1.7 million people were in Emergency (IPC Phase 4) and 47,000 were in Catastrophe (IPC Phase 5). These estimates were in the presence of ongoing humanitarian assistance.

Past data indicates food security outcomes do improve notably during the harvesting period, and food security improvements are expected through the end of 2018. However, even with these improvements, many households will continue to

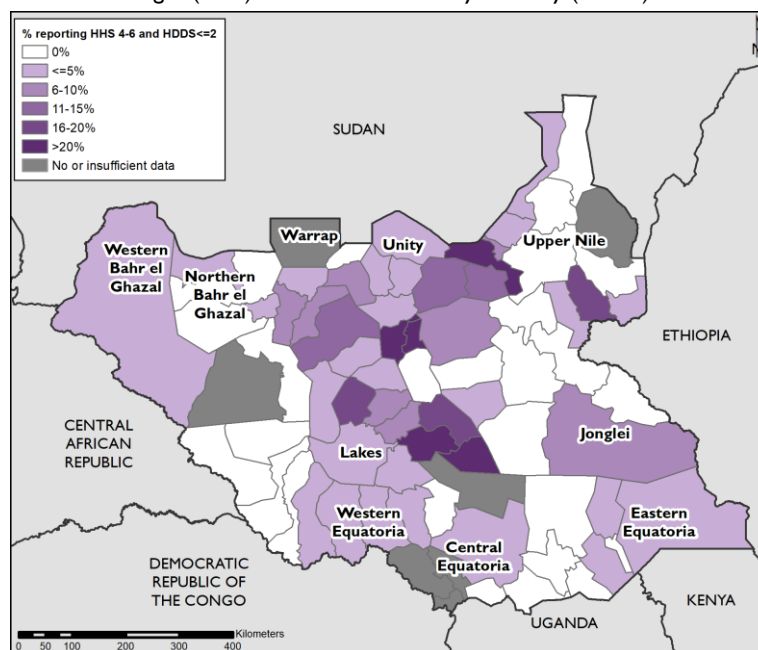
face Crisis (IPC Phase 3) or worse outcomes, and some households did not harvest and will not share in the improvements. Extreme food security outcomes, and associated excess mortality, is expected in South Sudan even during the 2018/19 harvesting period, and outcomes on par with the 2018 lean season are expected at the end of the projection period, which marks the 2019 lean season. The South Sudan IPC TWG estimated 4-5 million people will be in Crisis (IPC Phase 3) or worse through March 2019, including 26,000-36,000 in Catastrophe (IPC Phase 5). These estimates are also in the presence of planned humanitarian food assistance.

FEWS NET estimates this number will further increase throughout May 2019 as the lean season progresses. Humanitarian food assistance is likely to reach 2-3 million people per month throughout these time periods. However, the reach is still far below the estimated need. In the complete absence of assistance FEWS NET estimates around 6.5 million people would be in Crisis (IPC Phase 3) or worse by the end of the projection period, and Famine (IPC Phase 5) would be likely should there be a protracted absence of assistance.

Nearly five years of conflict across the country has eroded the capacity of many households to meet their basic food needs. Despite the signing of the most recent peace deal on September 12, conflict persists across the country. It is expected that conflict will continue in South Sudan, though there is a possibility of lower levels of conflict should the peace deal be successfully implemented. Past trends indicate that food security can deteriorate rapidly when conflict significantly restricts the movement of affected households and humanitarian access. Given the volatile nature of conflict in South Sudan, the areas of greatest concern can shift quickly. For this reason, although central Unity and Greater Baggari are of high concern currently, new areas of concern are possible throughout the projection period.

Humanitarian food assistance well above current planned and funded levels, complemented by unhindered access, is needed urgently and throughout the projection period to prevent the loss of lives and further erosion of livelihoods. An end to the conflict by all parties, and full implementation of the signed peace deal, is ultimately needed to end the persistent risk of Famine (IPC Phase 5) in South Sudan.

Figure 1. Percentage of surveyed households reporting both severe household hunger (HHS) and household dietary diversity (HDDS) <=2.



Source: FSNMS round 22 data

² Five counties were not analyzed due to insufficient data driven by access constraints to data collection: Maban, Terekeka, Lainya, Morobo, and Yei. Consequently, the analysis excluded the populations of these counties, which together equal roughly five percent of the population of South Sudan

NATIONAL OVERVIEW

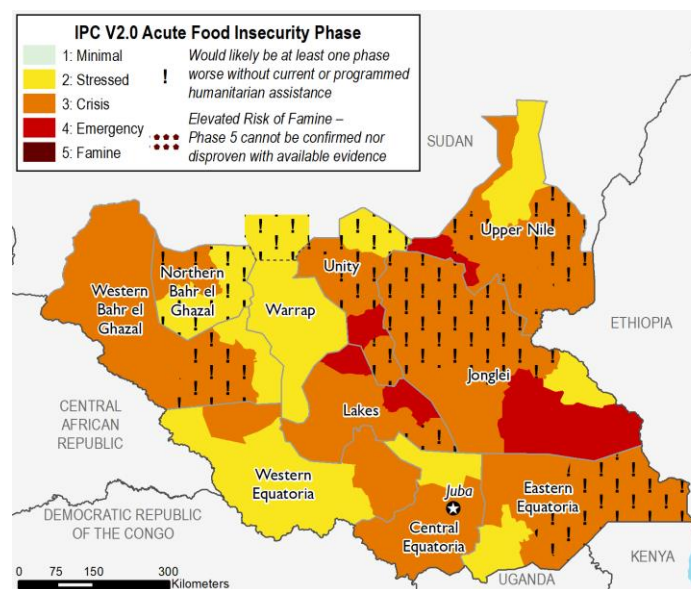
Current Situation

Extreme levels of acute food insecurity existed across South Sudan during the May-August 2018 lean season. According to the [September 2018 IPC analysis](#), an estimated 6.1 million people were in Crisis (IPC Phase 3) or worse at the peak of the lean season, though the need was likely slightly higher as five counties were not included in this analysis³. This estimate included 1.7 million people in Emergency (IPC Phase 4) and 47,000 people in Catastrophe (IPC Phase 5). However, these estimates were in the presence of food assistance and the total need was likely higher.

Food Security and Nutrition Monitoring System (FSNMS) data indicated that most households across the country had poor food consumption (FCS), medium or low dietary diversity (HDDS), and were experiencing moderate or severe hunger (HHS), indicative of widespread Crisis (IPC Phase 3) and Emergency (IPC Phase 4). An analysis of a convergence of these food consumption and livelihood coping outcomes at the household level showed that five percent or more of the population in 14 counties reported outcomes indicative of Catastrophe (IPC Phase 5). These data were complimented with ground information and corroborating evidence during the August IPC workshop and it was determined that households were likely in Catastrophe (IPC Phase 5) at the peak of the lean season in seven counties: Leer, Mayendit, Yirol East, Yirol West, Canal/Pigi, Panyikang, and Greater Baggari of Wau. In Gogrial East, Gogrial West, Awerial, Rumbek East, Mayom, and Fagak, based ground information analysts determined Catastrophe (IPC Phase 5) outcomes were determined unlikely.

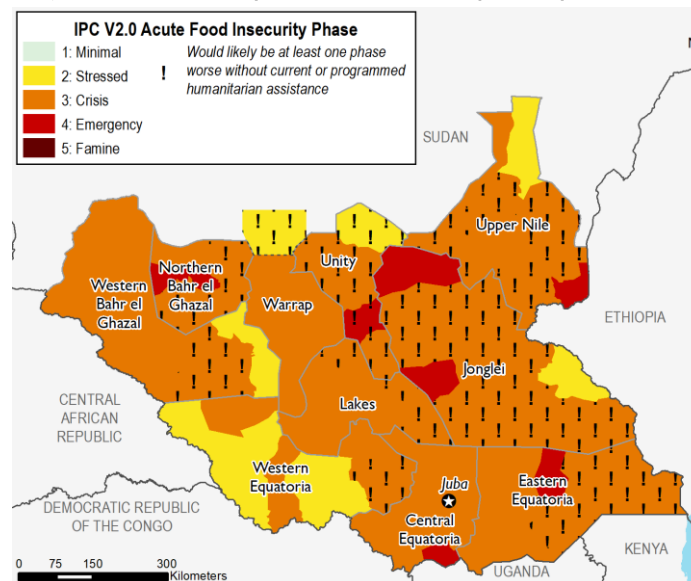
Of the 21 SMART surveys conducted between May and August, 15 counties reported a 'Critical' level (15.0-29.9%) of Global Acute Malnutrition (GAM), as measured by Weight-for-Height z-score (WHZ), and six counties recorded a 'Serious' (10.0-14.9%) GAM (WHZ) prevalence. These results are indicative of Emergency (IPC Phase 4) and Crisis (IPC Phase 3), respectively. It is expected these outcomes are mainly driven by lack of food or inadequate access to food, and poor access to safe water and nutrition and health services. In most areas, the non-trauma crude death rate (CDR) was indicative of Minimal (IPC Phase 1) or Stressed (IPC Phase 2), though in five counties the CDR was indicative of Crisis (IPC Phase 3) or Emergency (IPC Phase 4). Most SMART surveys also collected data on food consumption and the results of HHS and FCS also point to Crisis (IPC Phase 3) or worse outcomes, with some households across regions reporting outcomes indicative of Catastrophe (IPC Phase 5) (Table 1).

Projected food security outcomes, October 2018 to January 2019



Source: FEWS NET

Projected food security outcomes, February to May 2019



Source: FEWS NET

³ Five counties were not analyzed due to insufficient data driven by access constraints to data collection: Maban, Terekeka, Lainya, Morobo, and Yei. Consequently, the analysis excluded the populations of these counties, which together equal roughly five percent of the population of South Sudan

Relative to the 2017 lean season, the 2018 lean season was broadly similar, though there were some areas, such as Jonglei, where data indicated improvement, while other areas, such as central Unity, showed deterioration. The above-mentioned convergence analysis found that, overall, the size and severity of food insecurity during the 2018 lean season was slightly lower than that of 2017. Given that the number of conflict events was lower during the 2018 lean season and humanitarian food assistance deliveries were slightly higher, it is possible outcomes during the 2018 lean season were in fact slightly less severe. However, it should be noted that several of the most conflict-affected counties were not accessed during 2018 data collection, which may have positively skewed the results. Nutrition and mortality data indicated similar outcomes during the 2017 and 2018 lean seasons. Of high concern during the lean season were counties with populations in Catastrophe (IPC Phase 5), as well as counties where data indicated further deterioration beyond already extreme outcomes would be likely in the absence of assistance. Analyses on these counties is further detailed in the area of concern sections.

Despite the signing of the revitalized peace agreement on September 12 by the Government and opposition, conflict persists across much of the country. The conflict has continued for nearly five years and is the key driver of acute food insecurity in the country, causing significant disruption to normal livelihoods and the economy and displacing millions of people. An estimated 1.84 million people remain internally displaced, and 2.47 million people have fled the country.

Table I. Nutrition, mortality, and food consumption outcomes from SMART surveys, May – September 2018

State	County	GAM (WHZ)	SAM (WHZ)	Non-trauma CDR	FCS	HHS
EES	Kapoeta East	10.4% (7.5-14.3)	1.8% (0.8-3.8)	1.15	23% Borderline; 14% Poor	82% moderate; 1% Severe (0% HHS 5-6)
Lakes	Rumbek North	10.1% (N/A)	2.2% (N/A)	0.48	28% Borderline; 57% Poor	35% Moderate; 4% Severe (2% HHS 5-6)
	Yirol East	22.6% (19.3-26.3)	6.5% (4.8-8.6)	0.23	8% Borderline; 89% Poor	79% moderate; 20% Severe (20% HHS 5-6)
UNS	Maban	10.2% (7.7-13.5)	2.8% (1.6-5.0)	0.20	N/A	N/A
	Renk	26.9% (22.7-31.6)	6.4% (4.8-8.6)	1.26	38% Borderline; 19% Poor	53% Moderate; 0% Severe (0% HHS 5-6)
	Malakal	18.5% (15.1-22.4)	3.0% (1.9-4.9)	0.45	25% Borderline; 69% Poor	85% Moderate; 8% Severe (3% HHS 5-6)
Warrap	Tonj North	11.6% (9.0-14.9)	3.1% (1.8-5.3)	0.12	13% Borderline; 32% Poor	50% Moderate; 25% Severe (20% HHS 5-6)
	Gogrial East	18.0% (14.0-22.9)	2.8% (1.7-4.7)	0.34	15% Borderline; 33% Poor	54% Moderate; 12% Severe (9% HHS 5-6)
	Gogrial West	26.6% (21.0-33.0)	5.4% (3.3-8.7)	0.37	23% Borderline; 29% Poor	60% Moderate; 12% Severe (9% HHS 5-6)
Unity	Leer	12.2% (9.2-16.0)	3.0% (1.7-5.1)	0.47	2% Borderline; 98% Poor	62% Moderate; 33% Severe (23% HHS 5-6)
	Abiemn-hom	20.6% (17.2-24.5)	5.8% (4.3-7.5)	0.19	38% Borderline; 32% Poor	43% Moderate; 0% Severe (0% HHS 5-6)
	Guit	17.8% (N/A)	3.8% (N/A)	N/A	24% Borderline; 26% Poor	82% Moderate; 9% Severe (4% HHS 5-6)
	Mayom	18.3% (15.3-21.6)	3.2% (2.1-4.9)	1.05	32% Borderline; 62% Poor	64% Moderate; 9% Severe (9% HHS 5-6)
	Pariang	19.6% (15.5-24.5)	4.0% (2.6-6.1)	0.37	22% Borderline; 5% Poor	58% Moderate; 1% Severe (0% HHS 5-6)
	Rukbona	17.8% (15.2-20.7)	3.8% (2.6-5.6)	0.85	37% Borderline; 15% Poor	84% Moderate; 6% Severe (1% HHS 5-6)
WBeG	Wau	14.7% (11.1-19.3)	2.9% (1.7-4.7)	0.43	N/A	N/A
Jonglei	Akobo East	16.1% (12.4-20.5)	2.5% (1.3-4.7)	0.28	N/A	N/A
	Fangak	16.1% (12.1-21.1)	2.5% (1.5-4.4)	0.71	19% Borderline 56% Poor	64% Moderate; 36% Severe (10% HHS 5-6)
	Pibor	20.7% (CI: 17-25)	4.9% (3.1-7.4)	0.27	32% Borderline 57% Poor	86% Moderate; 13% Severe (13% HHS 5-6)
	Twic East	24.1% (19.8-28.9)	4.9% (2.7-8.5)	0.29	23% Borderline; 3% Poor	88% Moderate; 5% Severe (2% HHS 5-6)
NBeG	Aweil East	21.0% (17.4-25.1)	4.9% (3.2-7.3)	0.49	23% Borderline 40% Poor	N/A

Source: ACF - Duk, Gogrial West, Aweil East; CARE- Abiemnhoem, Rukbona, Guit, Pariang, Mayom; Tearfund- Twic East; WVI- Gogrial East, Tonj North; Help – Yirol East; CUAMM – Yirol West; IMC – Wau, Pibor, Malakal, Akobo; SCI- Kapoeta East; IRC- Maban; AAH- Fangak; Medair- Leer, Renk

The number of conflict incidents observed in 2018, as recorded by ACLED, were lower than in 2016/17; however, the number of incidents in October indicates conflict has recently increased. Typically conflict increases seasonally during this time period, though the increase observed in October 2018 was greater than usual (Figure 2). The recent increase is driven heavily by violence in **Wau** of Western Bahr el Ghazal and **Yei** of Central Equatoria. In addition, conflict has continued in **Koch** of Unity and attacks have occurred along Wau-Raga, Juba-Mundri-Yambio, Juba-Bor, Torit-Juba, and Kaya-Yei roads. Inter-communal conflict persists in Tonj South of Warrap, Cueibet of Lakes, Mundri of Western Equatoria, and **Terekeka** of Central Equatoria.

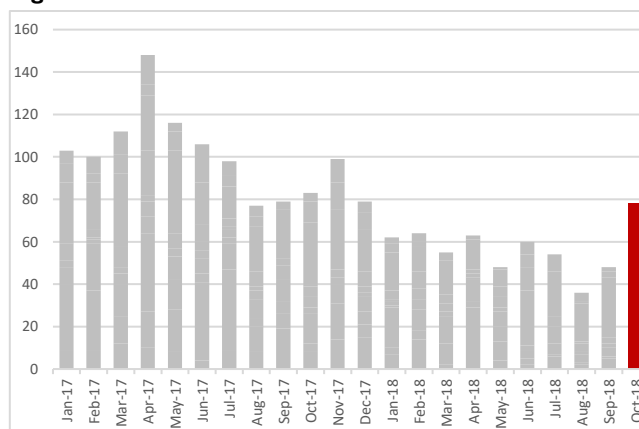
The conflict continues to disrupt normal market functioning and trade flows (Figure 3). Although markets are operational in state capitals, trade flows to rural areas remain very low and the high cost of operation, including illegal taxation, theft, risk of harm, and poor road conditions, drives extremely high prices and restricts many households' access to markets.

South Sudan's macroeconomic conditions remain very poor. The SSP appreciated significantly from 300 SSP/USD in mid-2018 to roughly 170 SSP/USD in June/July following an injection of USD into the country, though the currency has depreciated since July and is now trading at 213 SSP/USD. With the resumption of production in Toma South oil fields of Rubkona in Unity State, oil production increased from around 130,000 barrels per day (bpd) to 165,000 bpd. The increased oil production has not led to currency appreciation to date, though it is unclear if this is due to the fact that reserves are not yet available, or if total earnings will have a negligible impact on the value of the SSP. However, fuel prices at most stations in Juba have reduced slightly, from 180 SSP/liter in September to 175 SSP/liter in October.

In state capitals, the price of sorghum is around 110-140 SSP according to data from WFP and the Crop and Livestock Market Information System, though higher prices are observed in rural markets. Staple food prices have declined somewhat from mid-2018, though prices in most markets in October are still above the same time last year. Similar to trends observed in Juba (Figure 4), staple food prices in many areas of the country continued to increase year-on-year, though current increases are much lower than those observed in 2016/17. Staple food prices remain over ten-fold higher than pre-crisis levels.

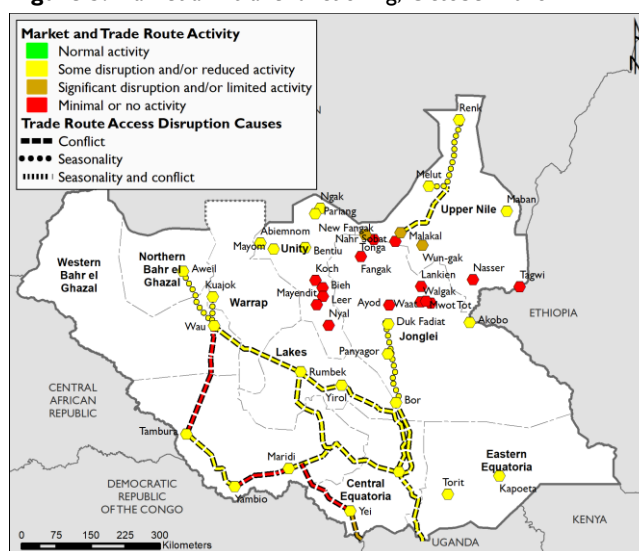
The June to September main rainy season in northern areas and second rainy season in Greater Equatoria started on time in most areas, though rainfall was erratically distributed over space and time in several areas. Overall, rainfall was below average in large parts of Lakes, Jonglei, Eastern Equatoria, and Central Equatoria (Figure 5). Although rainfall was overall average in Northern Bahr el Ghazal, Western Bahr el Ghazal,

Figure 2. Number of conflict events in South Sudan



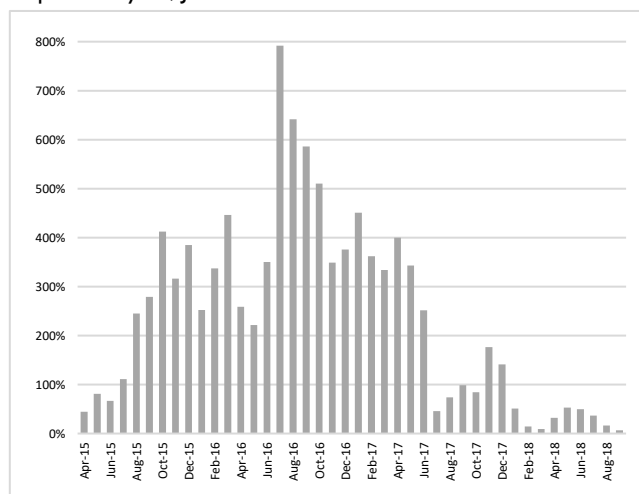
Source: ACLED data

Figure 3. Market and trade functioning, October 2018



Source: FEWS NET

Figure 4. Monthly percent change in price of sorghum compared to previous year, Juba



Source: WFP data

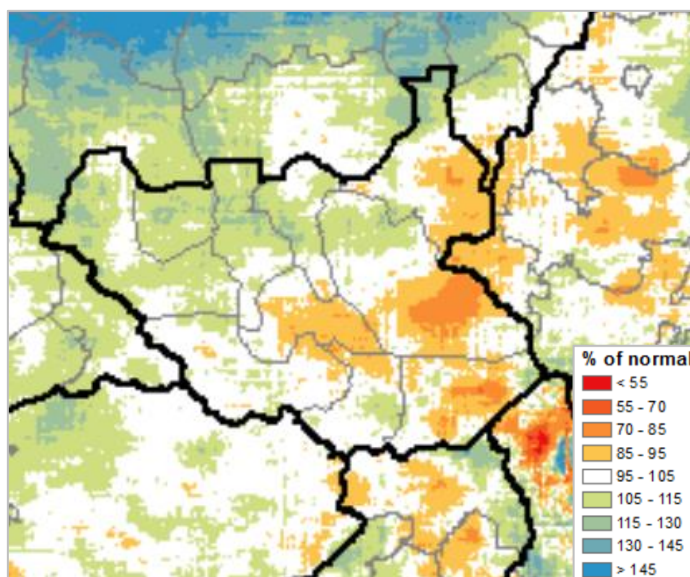
and Unity, dry spells were reported in June/July. Above-average late season rainfall largely made up for early season deficits; however, these dry spells were reported as a key barrier to cultivation by a significant number of households during July-September FSNMS data collection. Insecurity and Fall Armyworm (FAW) were also noted as main constraints. Overall crop losses due to FAW were considered slight to moderate, though some areas reported more significant crop damage.

In October, the harvesting of second season crops in Greater Equatoria and main season crops in Greater Bahr el Ghazal and Greater Upper Nile is ongoing. Although many households are harvesting, FSNMS data and rapid assessments conducted by FEWS NET and partners indicate that there are households in many areas who are unlikely to harvest due to conflict-related disruptions and displacement away from fields.

Humanitarian food assistance continues to play a key role in reducing the occurrence of extreme food insecurity. The majority of assistance is delivered in Greater Upper Nile, though several areas of Greater Bahr el Ghazal and Greater Equatoria are also receiving large-scale food assistance. Between July and September, general food distributions (GFD) were delivered to roughly two million people per month. Some areas receive assistance once every two or three months. It is therefore likely the beneficiaries reached each month were not the same two million people, and the total number of people reached throughout South Sudan was higher than two million. Monthly rations were between 33 and 100 percent of kilocalorie needs. This assistance is mitigating food consumption gaps for many, though it is reaching far less than the estimated population in need. For the past two years, roughly 20 percent of the population was reached with GFD each month. However, the SS TWG has estimated that roughly 40-60 percent of the population was still in need in the presence of this assistance. Throughout 2018, similar to recent years, conflict at times inhibited the delivery of humanitarian assistance, most notably in central Unity, Raga and Greater Baggari of Wau of Western Bahr Ghazal, and many areas of Central Equatoria.

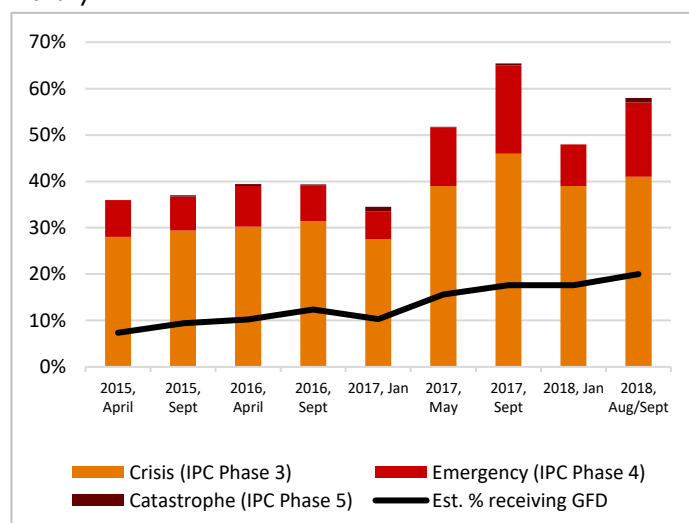
By October, it is expected that food consumption has improved somewhat with the ongoing harvest and increased availability of fish and wild foods following the rainy season, and the number of people facing food consumption gaps or engaging in negative livelihood coping to meet their basic needs as declined. However it is still estimated that between 4-5 million people are in need of assistance above needs already met by ongoing humanitarian food assistance. It is expected that households are still likely in Catastrophe in in Leer, Mayendit, Greater Baggari of Wau, Pigi/Canal, Yirol East, Yirol West, and Panyikang, and that some households have now deteriorated to Catastrophe (IPC Phase 5) in Pibor.

Figure 5. Seasonal rainfall accumulation as a percent of normal, June to September 2018



Source: FEWS NET/USGS

Figure 6. Severely food insecure population in the presence of assistance and percentage of the population receiving GFD on average monthly



Source: IPC TWG, WFP

Assumptions

The October 2018 to May 2019 most likely scenario is based on the following national-level assumptions:

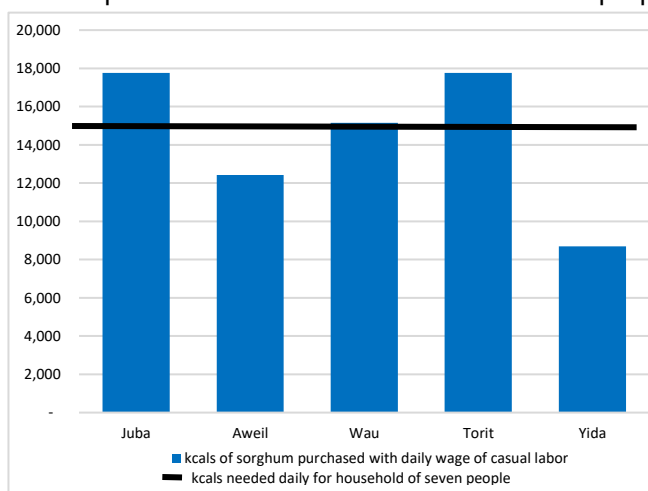
- Given the difficulty in projecting the evolution of conflict, this scenario assumes a continuation of conflict similar to recent years. It should be noted though that, given the [revitalized peace agreement](#), a decline in conflict is possible with successful implementation, and this best-case scenario is noted in Table 2.
- The 2018/19 harvest is expected to be similar to last year but lower than pre-crisis levels, based on data from FSNMS, CFSAM, and rapid assessments. Many households are likely to harvest; however, conflict-related disruptions, low household assets, prolonged dry spells, and the impacts of FAW and other pests will result in little to no harvests for some households. Among those who harvested, it is expected household food stocks will last 2-4 months in Greater Upper Nile, 2-5 months in Greater Bahr el Ghazal, and 3-7 months in Greater Equatoria.
- Based on forecasts from NOAA and USGS, the remainder of the August to November second rainy season is forecast to be average, and, despite current rainfall deficits over southern South Sudan, total seasonal rainfall is still expected to be average. The March to May 2019 first rainy season in Greater Equatoria is forecast to be average.
- With the timely start of the March to May rainy season, most households are likely to plant first-season crops in Greater Equatoria in March/April, though some lack productive assets and will be unable to plant. Given the likelihood of some returnees, the area cultivated will be slightly higher than 2018, but below pre-crisis levels as conflict is still expected to disrupt cultivation.
- Fish and wild foods are expected to be available throughout the outlook period, but the availability will decline seasonally during the November to April/May dry season. Overall access to these food sources will at times be limited due to conflict.
- Imports of sorghum from Sudan will be slightly higher than last year given the expectation of a near average harvest in Sudan and likely formal border opening; however, volumes traded will remain below pre-crisis levels. Imports of maize from Uganda will likely be higher than in 2017 and early 2018, as conflict along the key trade route has been somewhat lower. However, incidences of road ambushes are still likely and import volumes are expected to be lower than pre-crisis levels. Internal trade and market functioning are expected to remain similarly constrained by occasional conflict.
- Based on recent increases in [oil production](#) and plans to increase production further, it is likely total oil production will be greater than 165,000 bpd, increasing the country's oil exports and foreign currency reserves to levels higher than last year. As a result, the value of the SSP on the parallel market is expected to be between 200 and 250 SSP/USD.
- Based on FEWS NET's integrated price projections, it is expected that staple food prices will remain 8 to 10 times higher than pre-crisis levels and well above the five-year average. However, prices in Juba and Aweil are likely to be lower in 2019 than prices observed in 2018. In Wau, prices in 2019 are expected to remain higher than 2018 prices.
- According to WFP's Operation Plan for November 2018 through May 2019, general food distributions are expected to reach 2.5 to 3.5 million people per month. The Operation Plan through March 2019 is [49 percent funded](#) and based on current funding and past trends, further funding is likely and it is expected the majority of planned assistance through May 2019 will be delivered. Assistance delivery is likely to be occasionally disrupted in conflict-affected areas.

Most Likely Food Security Outcomes

Food security is expected to remain largely stable through December 2018 with the availability of the harvest and seasonally high availability of wild foods and fish, though access to wild foods will be periodically limited by insecurity in conflict-affected areas, and some households did not harvest and will not share in improvements. Levels of acute malnutrition are expected to improve to 'Serious' GAM (WHZ) of 10.0-14.9 percent in some areas. Relative to the 2018 lean season, a higher number of households will face no acute food insecurity or Stressed (IPC Phase 2) outcomes during this time period, though Crisis (IPC Phase 3) and Emergency (IPC Phases 4) will persist throughout the country. Humanitarian food assistance is expected to prevent more extreme outcomes, in particular in Greater Upper Nile and parts of Greater Bahr el Ghazal and Eastern Equatoria. In Leer and Mayendit, Greater Baggari of Wau, Pigi/Canal, Pibor, and Panyikang, some households are expected to be in Catastrophe (IPC Phase 5). These households will face extreme difficulty meeting their basic food needs as they have little or no harvests, lack typical sources of income, will likely face constraints to regularly accessing fish and wild foods, and conflict is expected to occasionally disrupt the delivery of humanitarian food assistance to populations in need in these areas.

Food security is expected to deteriorate through May 2019, and, similar to recent years, the lean season is expected to start by January 2019 with the early depletion of household food stocks. More widespread Crisis (IPC Phase 3) and Emergency (IPC Phase 4) outcomes are expected. In pre-crisis years, many households would rely on market purchases to meet a large portion of their food needs during the lean season. Given the likely persistence of extremely high food prices and low wages, the daily wage rate will remain insufficient to purchase a household's daily minimal kilocalorie needs in many areas (Figure 7); in some areas it will be minimally sufficient though it is unlikely households will access labor daily. Although the peak of the lean season is typically July/August, the severity of food insecurity is also expected to increase notably towards the end of the dry season (April/May) when the availability of wild foods and fish are seasonally low. Levels of acute malnutrition are likely to deteriorate to 'Critical' (15.0-29.9%) due to both the decline in food availability and seasonal increase in disease prevalence, and some areas may record a 'Very Critical' GAM (WHZ) prevalence ($\geq 30\%$). Catastrophe (IPC Phase 5) outcomes are also expected during this time in the above-mentioned counties of greatest concern with the exception of Panyikang where a recent decline in conflict is expected to enable better household movement and higher trade flows than were observed at the peak of the lean season. Humanitarian food assistance is expected to continue preventing more extreme outcomes in many areas. Given that severity of outcomes in central Unity at the peak of the lean season, and the expectation of little to no harvests among households of greatest concern, Famine (IPC Phase 5) is likely should conflict restrict the delivery of humanitarian food assistance throughout the projection period.

Figure 7. Kilocalories that can be purchased with daily wage rate compared to kilocalorie needs of a household of 7 people



Source: WFP data

Broadly speaking, the risk of Famine (IPC Phase 5) will persist in South Sudan throughout the outlook period, during both the post-harvest period and start of the 2019 lean season. In addition to central Unity, Greater Baggari is of extreme concern. Food security outcomes are already extreme and should ongoing conflict continue to restrict household movement and humanitarian access, it is expected an increased number of households will be in Catastrophe (IPC Phase 5) and the risk of Famine (IPC Phase 5) will increase. Throughout the county, the risk of Famine (IPC Phase 5) is greatest when conflict restricts household movement in toward typical food sources and humanitarian assistance deliveries. Significant deterioration in food security and nutrition outcomes due to these factors has occurred several times over the past few years in central Unity, Greater Baggari, and areas of Jonglei. Given the volatile nature of the conflict, there remains difficulty projecting the areas where households and humanitarians are likely to face these restrictions. Therefore, Famine (IPC Phase 5) would be likely in several areas in the event conflict prevented the delivery of food assistance and household movement for an extended period of time.

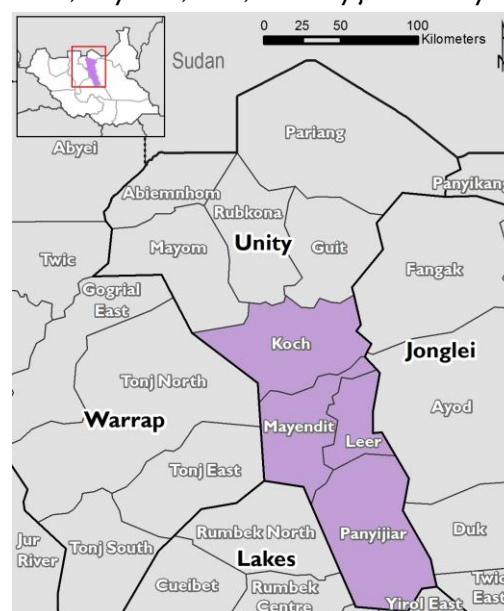
AREAS OF CONCERN

Unity State: Leer, Mayendit, Koch, and Panyijiar (Figure 8)

Current Situation

Persistent insecurity throughout central Unity has driven high levels of acute food insecurity since the onset of conflict in December 2013. Recent fighting from April to June 2018 in Leer and Mayendit, though, was among the most violent time periods of the conflict and led to death, displacement, and disruption of livelihoods and assistance deliveries. Many households faced

Figure 8. Area of concern reference map, Koch, Mayendit, Leer, and Panyijiar of Unity



Source: FEWS NET

severe food shortages during the lean season and these counties had some of the most extreme outcomes reported across the country.

In Koch, recent conflict has been somewhat less severe, but still of high concern. Heavy fighting in April led to the evacuation of humanitarians for a short period. In late August, armed clashes forced the displacement of some households from Pakur and Dhuar to Jaak and Petpet. In addition, [cattle raiding](#) remains a frequent hazard. Although security is relatively better in Panyijiar, this county remains of concern given the frequent arrival of displaced populations from central Unity. Currently, more than 70,000 IDPs are hosted in Panyijiar.

Following the outbreak of renewed conflict in April, humanitarian access was severely limited and GFD deliveries were halted. In Leer, the last delivery had occurred in January and a delivery was due in April, but was disrupted by conflict; therefore, no assistance was delivered between February and July. In Mayendit, between 10,000 and 35,000 beneficiaries were reached monthly in 2018 until fighting disrupted May and June deliveries. In Koch, no assistance was delivered in July, though large-scale assistance was delivered in August and September. In Panyijiar, GFD was delivered fairly regularly throughout the lean season.

In Panyijiar and Koch, according to FSNMS lean season data, most households reported poor food consumption, low dietary diversity, and moderate hunger (Figure 9). In Koch, nutrition data from FSNMS found a 'Serious' GAM prevalence of 13.2 percent. In Panyijiar, the FSNMS recorded a GAM (WHZ) prevalence of 19.6 percent. These food consumption and nutrition results are indicative of Crisis (IPC Phase 3) and Emergency (IPC Phase 4).

In Leer and Mayendit, food consumption outcomes indicated widespread Emergency (IPC Phase 4) with households in Catastrophe (IPC Phase 5). Households expected to be in Catastrophe (IPC Phase 5) are those who reported a poor FCS, engaging in consumption-based coping per rCSI, and an extreme lack of food associated with an HHS of 5 or 6. When data was collected in August, the first round of GFD since early 2018 had been distributed to Leer by ICRC but did not reach northern areas of the county. Food assistance had also not reached many areas of northern Mayendit at the time of data collection. It was in these locations that households reported outcomes indicative of Catastrophe (IPC Phase 5). The August FSNMS recorded a GAM (WHZ) of 13.7 percent (10.5-17.8) in Leer, though the results of GAM (MUAC), at 12.0 percent (7.4-18.8), and a July/August mass screening MUAC of 18 percent indicated the prevalence of acute malnutrition may have been higher. In Mayendit, the FSNMS recorded a GAM (WHZ) of 12.6% (4.5-30.4), indicative of Crisis (IPC Phase 3), though with a wide confidence interval. Admissions for outpatient therapeutic program and severe malnutrition with complications also spiked, though, in both counties during the time of increased conflict. This trend is in line with admissions recorded during other times of high conflict, restricted humanitarian access, and extreme food insecurity (Figure 10).

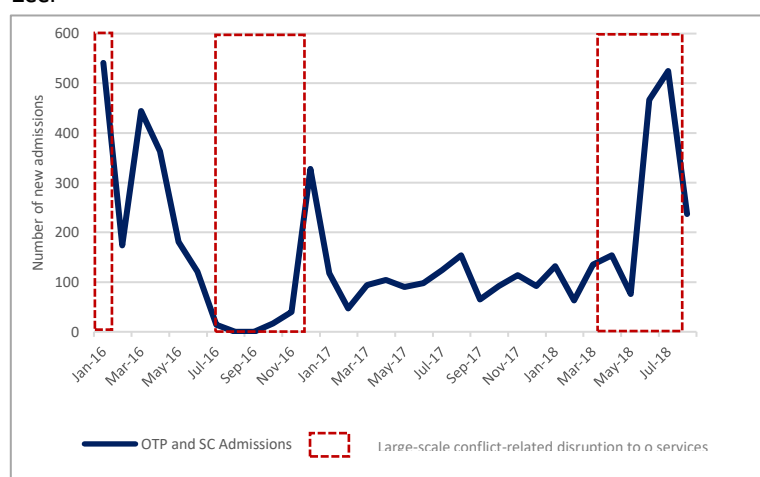
Since the peak of the lean season, conditions have improved, though overall access to food remains very low. New rounds of [fighting](#) in late September in Leer led to evacuation of WFP's Rapid Response Teams, though humanitarians returned in late September and food assistance deliveries have since resumed. Similarly, conflict in [mid-October](#) in Koch severely disrupted the delivery of assistance, though humanitarians have since regained access.

Figure 9. Food consumption outcomes in central Unity, July-Aug 2018

	FCS	HHS	HDDS	rCSI
Leer	2% borderline 98% poor	62% moderate 32% severe (22% HHS 5 or 6)	79% <=2 food groups	49% >=19
Mayendit	13% borderline 84% poor	43% moderate 37% severe (33% HHS 5 or 6)	83% <=2 food groups	39% >=19
Koch	30% borderline 35% poor	83% moderate 7% severe (0% HHS 5 or 6)	38% <=2 food groups	50% >=19
Panyijiar	37% borderline 33% poor	91% moderate 0% severe	62% <=2 food groups	28% >=19

Source: FSNMS Round 22

Figure 10. New monthly admissions for outpatient therapeutic program (OTP) and severe acute malnutrition with complications (SC), Leer



Source: FSNMS Round 22

Market functioning and trade flows in Leer and northern Mayendit remain very low after the escalation of conflict forced most traders to flee to southern Mayendit. Key informant information indicates that some traders are gradually returning, though insecurity along trade routes persists, keeping volumes traded low. In Paap market of Adok port in Leer, the retail price of a kg of sorghum decreased from 350 SSP in August to 220 SSP in September due to increased supplies from Juba, Panyijiar, and Fangak. Similarly, in Mayendit, the retail price of a kg of sorghum reduced from 300 SSP in August to 260 SSP in September. In Koch, insecurity and cattle-raiding continue to disrupt the consistent inflow of goods from Bentiu. In Panyijiar, market and trade functionality is relatively better, though ongoing insecurity and [boat ambushes](#) also lead to low supplies and high prices.

The ownership of livestock, a typical source of food and income prior to the conflict, is increasingly scarce. In Leer and Mayendit, under 20 percent of the population reported to the FSNMS owning any livestock at all and only 4-5 percent reported owning cattle. Most livestock losses occurred several years ago, though continued conflict and cattle raiding negatively impact households who still have livestock. Overall access to milk is limited in these counties, with over 90 percent reporting no access to milk from any source. In Panyijiar, 52 percent reported access to milk, and 48 percent in Koch, where livestock ownership is relatively higher, but still lower than pre-crisis levels.

Rainfall started normally in May and despite localized dry spells in June, total seasonal rainfall was near average and heavier than last year. Flooding occurred in some lowland areas of Mayendit. Ground information indicated access to land and inputs at the beginning of the season was better than in previous seasons, though conflict significantly disrupted the season in Leer

and Mayendit, and only some households re-planted in mid-June. Cultivation was better in Koch where nearly 90 percent of households reported cultivating at least 1-2 feddans and conflict did not lead to significant disruptions. In Panyijiar, over 50 percent of households stated they received seeds and tools and 94 percent reported cultivating an average of 2.3 feddans. It is expected that relative security also contributed to this higher level of planting. However, over 77 percent of households in Panyijiar reported that FAW negatively impacted crops.

It is expected that outcomes have improved somewhat with the harvest, though the situation remains fragile and outcomes are severe. In Koch improvement is also driven by an increase in the number of beneficiaries receiving humanitarian food assistance in recent months, whereas in Panyijiar the percentage of the population receiving food assistance has remained consistently high. Crisis (IPC Phase 3!) outcomes are likely in both counties. In Leer and Mayendit, with the resumption of GFD deliveries and the harvest available for some, it is expected that the population in Emergency (IPC Phase 4) and Catastrophe (IPC Phase 5) has declined. However, given limited other food sources and the likelihood that not all in need are reached by assistance given displacement and insecurity-related restrictions to free household movement, it is expected Emergency (IPC Phase 4) outcomes persist, with some households in Catastrophe (IPC Phase 5).

Assumptions

In addition to the national-level assumptions above, projected food security outcomes for the counties of concern in Unity State are based on the following assumptions:

- Fish will be available throughout the outlook period, though availability will be seasonally lower between January and April when households must access distant rivers to fish. Some will face challenges accessing a high volume of fish due to limited ownership of fishing equipment. Given security issues with traveling long distances, it is unlikely women will regularly fish and, as a result, female-headed households are unlikely to have regular access to fish between February and April.
- It is likely that trade and market functionality will remain minimal in the counties of concern, though traders who fled in mid-2018 are likely to return during the projection period and supplies will increase somewhat.
- Households who harvested are expected to have 1.5 months of cereal, though some households will not have any harvests.
- The availability of milk will decrease seasonally from January through April when livestock migrate to dry season growing areas. Many households are unlikely to consume livestock products or have income from the sale of livestock based on very low livestock ownership, though access to these food and income sources will be higher in Koch and Panyijiar.
- Water lilies are expected to be available from October-March. Other wild foods will also be available, though the availability of these wild foods will be seasonally low in November and December. The period of lowest food availability during the

outlook period is expected to be April-May, when water lilies, fish, and milk availability are very low and fruits, kundra, tameleka and other wild foods are the only food sources broadly available.

- According to WFP's Operational Plan for assistance between November 2018 and May 2019, over 50 percent of the population in each of the counties of concern is expected to be reached with GFD on average monthly, with the equivalent to 33-50 percent of their daily energy needs. In Panyijiar, planned and funded assistance is likely to be delivered as planned given relative stability. In Leer, Mayendit, and Koch, assistance is likely to be distributed somewhat regularly throughout the projection period following the return of the RRM teams, though periodic disruptions are likely based on ongoing insecurity and past trends.

Most Likely Food Security Outcomes

With the availability of harvests and increased fish and wild foods, food security will improve marginally in all counties of concern through the end of 2018. Despite this and the presence of large-scale humanitarian assistance, many households are expected to face large food consumption gaps. In Panyijiar, relatively better access to livestock products and harvests will lead to larger food security improvements for some; however, many IDPs rely heavily on humanitarian assistance and it is expected the likely delivery of this assistance will prevent more extreme outcomes and Panyijiar will remain in Crisis (IPC Phase 3!) throughout the projection period. In Koch, where planting and access to livestock are also relatively better, and humanitarian assistance is likely to be delivered somewhat regularly, Crisis (IPC Phase 3!) is also expected throughout the projection period.

In Leer and Mayendit, food security is expected to remain broadly similar even in the presence of humanitarian food assistance, given that many did not harvest, and the only other food sources are fish and wild foods. With the resumption of assistance, it is expected food consumption gaps will be smaller and food security will improve for many. However, given ongoing insecurity in the area that is likely to result in occasional disruptions to markets, wild foods, and fish, and given that the monthly ration size is only expected to cover 30-50 percent of energy needs and it is unlikely all households in need will be reached, Emergency (IPC Phase 4) outcomes are still likely. Should humanitarian food assistance be completely absent from Leer and Mayendit throughout the projection period, given the severity of outcomes already occurring and past trends that indicate outcomes can deteriorate to levels near or above Famine thresholds in the prolonged absence of assistance, Famine would be likely in the second projection period. For this reason, these counties are classified in Emergency (IPC Phase 4!) between February and May.

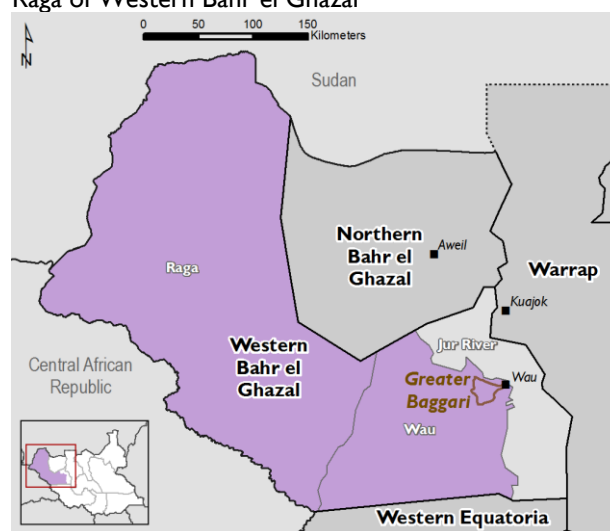
Western Bahr el Ghazal State: Wau and Raga (Figure 11)

Current Situation

The sub-area of Greater Baggari in Wau became an area of high concern in 2016 after protracted conflict led to significant displacement, culminating in extreme limitations to household movement and concerns of an elevated risk of Famine (IPC Phase 5) by mid to late 2017. Though there was relative calm in late 2017/early 2018, conflict again sharply increased in June 2018 (Figure 12) and persists through October, leading to further internal displacement of 31,000 people from Ngo dakalla, Farrajallah, Ngisa, Ngo vendego, and Mboro to Bazia, Bessilia, and unknown rural areas. Fighting between the Government and opposition forces in mid-October along the Wau-Mboro road led to further loss of lives, destruction of assets, and limited trade and movement between Greater Baggari and Wau town, and this is driving renewed concern of extreme food insecurity.

Although to a lesser extent, conflict also persists in Raga, where clashes in March and April 2018 resulted in displacement from Diem Zubeir and Mangayat to Kuru, Yabulu, Raga town, Tambura, Namutina, and Wau Protection Of Civilians Adjacent Areas; [REACH reported](#) the majority of the displaced were women and children. Conflict also disrupted cultivation, corroborated by FSNMS data in which 79 percent of households reported insecurity as main factor limiting cultivation.

Figure 11. Area of concern reference map, Wau and Raga of Western Bahr el Ghazal



Source: FEWS NET

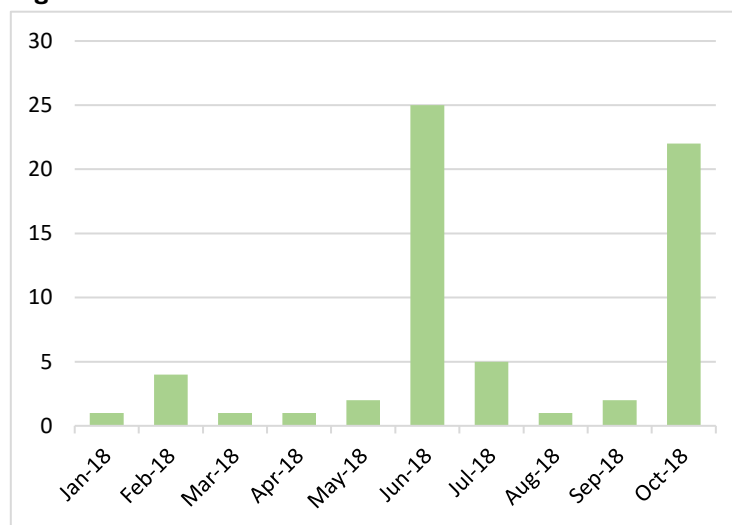
In both Wau and Raga, land preparation started in February and March and planting took place immediately after the early onset of the rains in April. However, rainfall was erratically distributed across space and time, confirmed by key informants who reported prolonged dry spells between May and June. The dry-spell affected the development of groundnuts, maize, and short-season varieties of sorghum in many areas, most significantly in Kuajena and Dam Zubeir. Conversely, some areas were affected by heavy late-season rainfall and floods were reported in July and August in Ringi, Erre, Momo Raga, Kuru, and Kapakange of Raga, destroying an estimated 7,868 feddans of crops. 77 percent of surveyed households in Wau reported facing difficulty cultivating due to insecurity and conflict, though due to access constraints FSNMS data was only collected among 36 households in Udo of Greater Baggari. With assistance from FAO of inputs to roughly 8,000 households, it is expected many households planted; however, many fields were left unattended or destroyed by armed forces, especially along Wau-Bazia-Mboro road.

Conflict also continues to limit normal markets and trade functioning in both Wau and Raga. In Raga, 78 percent households reported to the FSNMS the existence of a functioning market but with low staple food supplies; key informant information indicates that the main supply routes from Sudan through Warawar-Aweil and Gok Machar-Nyamlell are disrupted by insecurity and poor road conditions. In Wau, the retail price of a kg of sorghum remains nearly 400 hundred percent above the five-year average. It is expected poor households are earning small amounts of money through the sale of firewood and tea, though given extremely high prices households face difficulty accessing foods from markets. In many rural areas including Greater Baggari, physical access to markets is limited, and ground reports indicated households face restrictions moving toward Wau town.

In Raga, food frequency indicators (FCS and HDDS) pointed to Emergency (IPC Phase 4) outcomes while HHS, an experiential indicator, pointed to Crisis (IPC Phase 3) at the peak of the lean season. Between June and August, an estimated 7 percent of the population is being reached by GFD according to an analysis based on WFP distribution reports. It is expected that Crisis (IPC Phase 3) outcomes persist in October as households are consuming some harvests, though dry spells and conflict-related disruptions have resulted in lower than normal production.

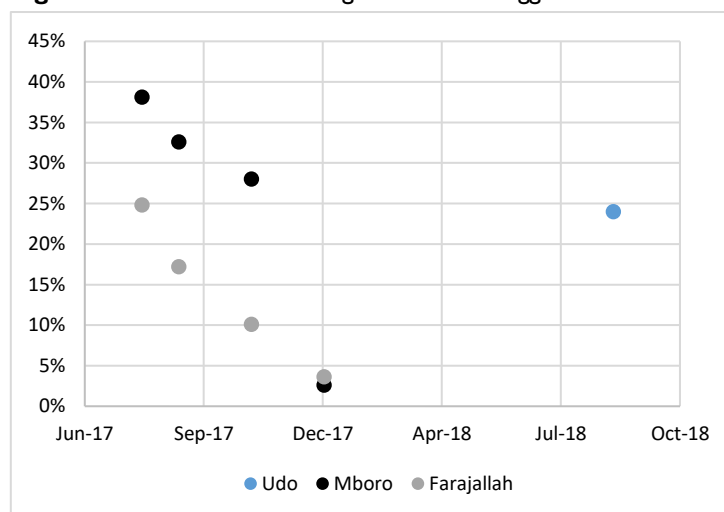
In Wau, households are located in Wau town and PoC, Greater Baggari, and to a lesser extent in rural areas outside of Greater Baggari. A SMART survey conducted by IMC in Wau town in July 2018 recorded a GAM (WHZ) prevalence of 14.7 percent (11.1-19.2), a non-trauma CDR of 0.41/10,000/day (0.29-0.74), and an U5DR of 0.7/10,000/day (0.20 - 2.34). The CDR is indicative of Minimal (IPC Phase 1) or Stressed (IPC Phase 2) outcomes, while the CDR is indicative of Stressed (IPC Phase 2) and the GAM (WHZ) is indicative of Crisis (IPC Phase 3). The SMART survey also collected food consumption data, and 33 percent of households reported borderline FCS and 27 percent reported moderate HHS both indicative of Crisis (IPC Phase 3). Among the 36 households surveyed in Greater Baggari, 82 percent of household reporting a poor FCS, 66 percent reported consuming 2 or less food groups, and 72 percent of household reporting moderate HHS, which are indicative of Crisis (IPC Phase 3) and

Figure 12. Conflict incidents in Wau



Source: ACLED

Figure 13. Nutrition screenings in Greater Baggari



Source: Nutrition working group

Emergency (IPC Phase 4) outcomes, though these data cannot be considered representative of Greater Baggari given the small sample size and lack of representative sampling. Limited information is available on the smaller population of households living in rural areas outside of Greater Baggari, though it is expected they face similar outcomes as those in Wau town.

In Greater Baggari, there was concern Famine (IPC Phase 5) may have occurred in mid to late 2017 when conflict significantly limited household movement and humanitarian access, and MUAC screenings showed extremely high malnutrition by proxy MUAC. These prevalences declined following regained humanitarian access that led to the provision of assistance and with the harvest. Although there were no screenings conducted between January and August 2018, it was expected malnutrition remained somewhat lower in the first half of 2018, in line with December 2017, given humanitarian access and relatively free household movement towards food sources. Following an uptick in violence in June, though (Figure 13), humanitarian access has been extremely limited. A 15-day ration was distributed to 6,500 beneficiaries in Mboro in May and 13,500 people were reached with GFD in Ngo Dakalla, Farajallah, and Ngisa in September, though during the September distribution, a MUAC screening of 1,649 children under five years of age reported a proxy GAM (MUAC) of 24 percent and SAM (MUAC) of 4.6 percent, indicating a likely sharp deterioration in outcomes. Since the September screening and GFD delivery, humanitarians have not been granted access to the area and it is expected outcomes remain extreme.

Assumptions

In addition to the national-level assumptions above, projected food security outcomes for the counties of concern in Western Bahr el Ghazal are based on the following assumptions:

- Intermittent clashes are still expected to continue in some parts of Raga and southern Wau including Greater Baggari areas throughout the scenario period intensity. Additional displacements within the counties and to the neighbouring counties of Northern Bahr-el Ghazal state and Jur River are likely.
- In both Wau and Raga stocks are expected to last for about three months; Many households in Greater Baggari, most notably in Ngo-Baggari, were forced to abandon their fields during the cultivation period and are unlikely to harvest or will harvest small amounts.
- Based on FEWS NET's integrated price projection for Wau, staple food prices are projected to follow seasonal trends, but be well above last year and five-year average, at around 100-150 SSP/kg. Market functioning and trade flows will also remain periodically disrupted by conflict.
- Income from the sale of firewood is likely to be limited due to insecurity-related restrictions to gathering these resources.
- According to the available WFP operational through May 2019, minimal to no GFD will be distributed to Raga until the second projection period, during which time an estimated 15 percent of the population will receive assistance. In Wau, WFP plans to reach between 30 and 40 percent of the total population with a third or half ration throughout the projection period. It is expected that a large proportion of this assistance will be delivered to the Wau PoC.
- It is unclear the amount of this assistance that is planned for Greater Baggari, though based on information from humanitarian partners, it is expected they will continue to seek access to Greater Baggari on a regular basis to deliver food assistance. It is expected that they will gain access at some point in late 2018/early 2019 to deliver assistance, though they are likely to face periodic access restrictions throughout the projection period.

Most Likely Food Security Outcomes

In Raga and areas of Wau, excluding Greater Baggari, food security is expected to improve slightly between October and December due to availability of some harvests and seasonally high wild foods and fish, though not all households will regularly benefit from these sources. Crisis (IPC Phase 3) is expected in Raga. In Wau town and PoC, where the majority of the population currently resides, it is expected humanitarian food assistance will prevent more extreme outcomes among beneficiaries and Crisis (IPC Phase 3!) will persist. By early 2019, households are likely to deplete food stocks and rely heavily on wild foods, hunting, and fishing, and although these sources will be insufficient to meet basic food needs, it is expected Crisis (IPC Phase 3) outcomes will continue in Raga, and with continued humanitarian food assistance in Wau, Crisis (IPC Phase 3!) is likely.

In Greater Baggari, Emergency (IPC phase 4) outcomes are expected to persist throughout the projection period. Some households were unable to cultivate and movement outside of Greater Baggari to collect wild foods and fish, or access resources in Wau PoC, is expected to remain volatile and at times not possible. Given persistent access constraints, there is limited information on current conditions and it is difficult to project how severe the situation is likely to be in the coming months, though based on the likely sharp increase in malnutrition in September and past trends of outcomes when assistance and access to food sources were severely limited, it is expected outcomes will further deteriorate. It is likely humanitarians will gain access at various points throughout the projection period and be able to deliver at least some assistance, though it is expected this will be irregular. Catastrophe (IPC Phase 5) outcomes are likely throughout the projection period. In the complete absence of assistance, it is possible Famine (IPC Phase 5) could occur.

Jonglei: Ayod, Nyirol, Fangak, Canal/Pigi, Duk, Uror, and Pibor Counties (Figure 14)

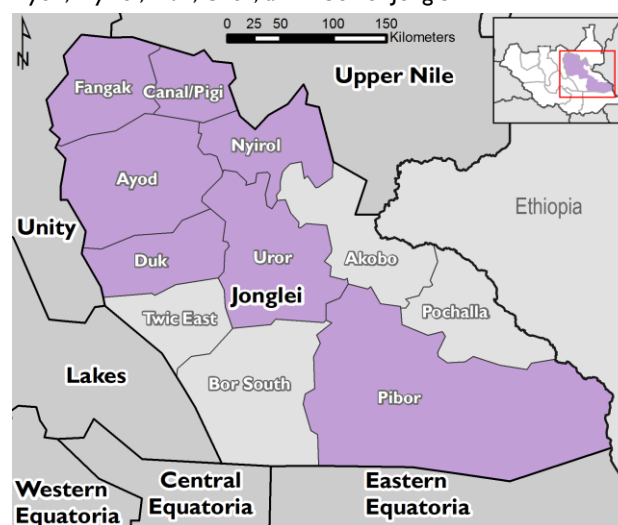
Current Situation

Crisis (IPC Phase 3) and worse outcomes persisted at the peak of the lean season. A SMART survey conducted in May 2018 in Fangak recorded GAM (WHZ) prevalence of 21.6 percent in Fangak; the FSNMS recorded a GAM (WHZ) of 34.8 percent. A GAM (WHZ) prevalence of 20.7 percent was recorded in Pibor, and in Twic East, a SMART survey conducted in May found a GAM (WHZ) prevalence of 24.1 percent (19.8-28.9). These results indicated Emergency (IPC Phase 4) or worse outcomes. In all the counties of concern, extreme food security outcomes existed during the peak of the lean season. In all counties, over 20 percent of the population had a poor food consumption score. In most counties, the majority of the population reported a moderate HHS, a notable improvement from the last lean season, though still indicative of Crisis (IPC Phase 3) outcomes. In Pigi/Canal, Pibor, and Fangak, over 10 percent of the population reported severe hunger, indicative of Emergency (IPC Phase 4).

Conflict persists in many areas of Jonglei and continues to affect poor households' access to food, income, and typical livelihood activities, and has resulted in large-scale displacement. Fighting between Government and opposition forces between March and June in Yuai, Pieri, and Motot of Uror, neighbouring Waat in Nyirol, and Walgak in Akobo East displaced many people and disrupted agricultural activities during the planting season. Recurrent fighting in Motot, Pieri, Waat, and Walgak also forced the displacement of an estimated 50,000 people from Karam Guich to Akobo East. Inter-communal conflict in Akobo and Ayod has also negatively impacted cultivation and trade flows. In Bor South and Nyirol, recurrent inter-communal fighting displaced over 8,700 people in Abii of Kolnyang and 5,000 people from Makolchai. In Poktap of Duk, cattle-raiding in October led to loss of life and livestock.

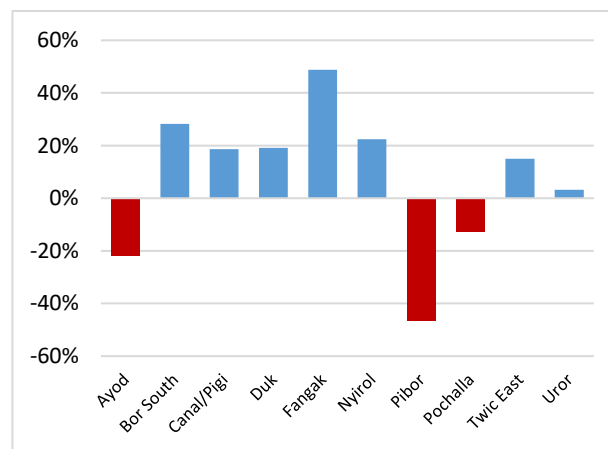
Although cattle raids and child abductions persist in 2018, key informant observation indicates lower incidence than last year. Additionally, the number of conflict incidents across Jonglei in 2018 according to ACLED was relatively lower than 2017. It is expected this is part of the reason a higher percentage of households had access to land and planted in 2018 compared to 2017 in most counties (Figure 15). However, the scale of planting was still low, with over 40 percent of the population in all counties reporting cultivating less than half a feddan.

Figure 14. Area of concern reference map, Fangak, Canal/Pigi, Ayod, Nyirol, Duk, Uror, and Pibor of Jonglei



Source: FEWS NET

Figure 15. Change in the percentage of households who reported planting for the 2018/19 season compared to the 2017/18 season



Source: FSNMS Round 22

was still low, with over 40 percent of the population in all

Dry spells remained the major challenge during the planting season in Canal/Pigi, Bor South, Duk, Nyirol, Twic East, and Uror. In addition, outbreak of Sudan diach (*quelea quelea*) was reported in all counties by the September 2018 IGAD Eastern Africa crop monitoring bulletin. Additionally, 40-90 percent of households across the counties of concern reported FAW infestation on their crops, and over 20 percent of households in Canal/Pigi, Fangak, and Pibor reported significant damage due to the pest.

Market functioning is very low or non-existent across the counties of concern. In Ayod, markets have not functioned since the outbreak of the conflict in December 2013 with the exception of the sale of small-scale goods such as sugar, salt, and soap. Markets are similarly limited or non-existent in Canal/Pigi and Pochalla. In Uror, Yuai market is functioning, but with limited commodities. In Nyirol, Lankien remains the only functioning market. Due to conflict in Duk, Fangak, Ayod, and Nyirol, insecurity along trade routes has negatively affected market supplies. The retail price of a kg of sorghum in Pibor, Bor South, and Akobo in September was roughly 500 percent higher than last year and 12-fold higher than five-year average.

Livestock remain a key source of food and income. Over 80 percent of households reported owning livestock in Fangak, Duk, Nyirol, and Pibor, though Ayod and Uror, under 50 percent of households owned livestock, and in Pigi/Canal where less than 6 percent reported owning livestock. Over 40 percent of households reported a large decrease in livestock ownership compared to the same time last year, due primarily to disease and cattle raiding.

Despite increased seasonal availability of fish, insecurity and lack of fishing gear limits the ability of households to fully engage in fishing. Access to fishing grounds is very limited in Nyirol, Ayod, and Uror due to insecurity, and was corroborated by FSNMS data in which only 1-2 percent of households reported fishing. Access to fish is relatively higher in the other counties.

It is expected most households are heavily dependent on wild foods to meet their basic food needs and increased consumption of wild foods may be indicative of continued coping: 30-75 percent of households in Pibor, Fangak, Nyirol, and Ayod reported in the FSNMS that at least one household member was consuming a new type of wild food they had not eaten before.

An analysis of WFP distribution data indicated that GFD likely reached over 25 percent of the population in all counties of concern except Pibor and Pigi/Canal. In Uror, a high percentage of the population was reached, though cut-off populations around Motot and Pulchul count not access distribution sites. Food consumption and nutrition data in Fangak indicated that extreme outcomes persisted even in the presence of large-scale humanitarian assistance; however, ground information indicated that the severity of outcomes was not likely as high as data may have indicated. It is still expected, though, that large food consumption gaps existed for many worst-affected households even in the presence of large-scale assistance in the county.

By October, with the harvest and increased availability of milk and wild foods, it is expected food security outcomes have improved somewhat, though large-scale humanitarian assistance continues to be a key factor in mitigating further deterioration. Crisis (IPC Phase 3) outcomes are likely in the counties of concern, except for Pigi/Canal, Fangak, and Pibor, where Emergency (IPC Phase 4) outcomes are likely.

Assumptions

In addition to the national-level assumptions above, projected food security outcomes for the counties of concern in Jonglei are based on the following assumptions:

- Intercommunal conflict and cattle raiding are expected throughout the scenario period, and the intensity likely to increase during the dry season between December 2018 and April 2019. This will restrict movement, and disrupt access to food and income sources, especially in Canal/Pigi, Fangak, Bor South, Pibor, Ayod and to some extent Akobo.
- Despite the slightly higher number of households planting, due to prolonged dry spells and crop losses from pests, the harvest will be similar to last year and below pre-crisis levels.
- Based on past trends and WFP's 2018 Operation Plan, general food distribution will likely reach over 50 percent of the population in Ayod, Fangak, Nyirol, and Uror. It is expected this planned assistance will be delivered through at least through May 2019, and likely to be delivered given relative calm. Assistance is planned to reach under 20 percent of the population in Duk, Pigi/Canal and Pibor. In all areas, assistance is expected to be between a quarter and half ration.

Most Likely Food Security Outcomes

Food security is expected to improve somewhat during the October to January period with the availability of the harvest. In addition to harvest, consumption of fish, and wild foods are also expected to reduce food consumption gaps that existed during the peak of the lean season. Crisis (IPC Phase 3!) is expected in Ayod, Duk, Uror, Nyirol and Fangak. Emergency (IPC Phase 4)

outcomes are expected to persist in Pibor and Pigi/Canal as households are expected to face large food consumption gaps due to significantly below average harvests as dry spell during the cropping season. In Canal/Pigi, where household with little or no harvest are also expected to be those with limited livestock ownership, Catastrophe (IPC Phase 5) outcomes are likely in the absence of higher levels of assistance. In Pibor, where large-scale assistance is planned, Catastrophe (IPC Phase 5) outcomes are still given that milk will be very low for many households, in particular during the dry season, and ongoing cattle raiding is expected to limit the delivery of assistance to some households in greatest need.

In all counties of concern, food security is expected to deteriorate between February and May due to multiple factors including depletion of food stocks in January, and likely inter-communal and cattle raiding. Households are expected to face food consumption gaps, and are likely to primarily rely on assistance, and consumption of fish, wild food and game meat expected to increase. However, access to livestock products will remain negligible for some households with few livestock and freedom of movement restrictions expected to persist. With these restrictions coupled with the above stated shocks, significant consumption gaps are expected to persist in all concern counties through May 2019. Crisis (IPC Phase 3!) outcomes expected to persist in Duk, Ayod, and Uror due to likely continued delivery of assistance in these counties given the relative calm. Emergency (Phase 4) outcomes are expected in Fangak, and Pigi/Canal, Pibor, and Nyirol, where planned assistance during this time period is relatively lower. In Pibor and Canal/Pigi, some households are expected to remain in Catastrophe (IPC Phase 5) due to the same reasons mentioned above.

EVENTS THAT MIGHT CHANGE THE OUTLOOK

Table 2. Possible events over the next eight months that could change the most-likely scenario.

Area	Event	Impact on food security outcomes
National	Implementation of the revitalized peace agreement	Efforts to implement the revitalized peace agreement signed in September by the government and opposition would improve security and allow households to slowly regain their livelihoods and return to their places of origin. It is expected that the incidence of conflict would decline and trade flows and humanitarian assistance could better reach households in need, and households would be able to move in search of food and pursue income-earning opportunities. Food insecurity would be less severe and the risk of Famine (IPC Phase 5) would decline.
National	Absence of humanitarian assistance	In the persistent absence of food assistance over a large area, Famine (IPC Phase 5) would be likely because this absence of assistance would remove a primary food source and would likely drive increased levels of conflict over remaining scarce resources. In turn, higher levels of conflict would increase movement restrictions, preventing households from accessing food from other sources. Given current food security outcomes and past conflict trends, areas of greatest concern include central and southern Unity, Jonglei, and Western Bahr el Ghazal. However, given the volatile nature of the conflict, and that food security can deteriorate rapidly among populations who face extreme movement restrictions, Famine (IPC Phase 5) remains possible in some areas of the country.